

BSc INFORMATION TECHNOLOGY



The Network is the Computer

The World Economic Forum's Future of Jobs Report 2018 predicts that a range of Information Technology related roles such as Data Analysts and Scientists, Software and Applications Developers, and Ecommerce and Social Media Specialists will increase in demand significantly.

With the move towards digitization globally and specifically in Ghana, there is a need for more people to participate in the digital economy. This presents opportunities for transforming the economy making us competitive in West Africa and allowing young people to find decent jobs or start their own businesses.

However, there is a substantial challenge. There is a huge skills gap. The only way to bridge the gap between the challenge and this immense opportunity is to train young people in these technologies.

Information Technology at ACity

This program will provide students the knowledge, skills and ability required to practice IT in this dynamic and fast-paced industry. Also, students will have the opportunity to take electives in these emerging areas and to build the necessary competencies required for practice and further studies. Finally, the syllabus has been designed to enable students to spot entrepreneurial opportunities in these areas to start their own business after graduation.

Top Careers in Information Technology

Graduates of this program will be able to select, create, apply, integrate, and administer computing technologies to meet the needs of users within societal and organizational contexts.

Possible career paths include:

- 🖱 Computer System Administrator
- 🖱 Network Administrator
- 🖱 I.T. Project Manager
- 🖱 Computer Analyst
- 🖱 Technical Writer
- 🖱 Webmaster

Why ACity?

Home to
students from over
23 countries
across Africa

A fully digitized
campus with the fastest
internet speed
in a Ghanaian university

70%
practical lessons

**Experienced
Faculty**
with Industry Expertise

Every **ACitizen** has
the opportunity to do a
work placement
every vacation

Student-To-Faculty Ratio
5:1

Houses a
Digital Makerspace
and a **Technology &
Entrepreneurial Center**
for **Creative freedom
& Innovation**

Our Unique Learning Pillars



Experiential Learning

Hands-on learning to prepare students to readily apply concepts, to easily integrate into the workspace.



Contextual Learning

Solving real grass-root problems to expose students to the local context and develop empathy towards the continent's progress.



Unified Learning

A project-based approach that combines concepts across courses to connect the dots and enable unified learning.



Extensional Learning

Arms students with a viable toolkit to help them confront real-life issues, they may not have encountered during their academic life, squarely.

Entry Requirements

WASSCE/NECO

Minimum C6 in 6 subjects including 3 core subjects (Maths and English mandatory) and 3 elective subjects

SSCE

Minimum D or a pass in 6 subjects including 3 core subjects (Maths and English mandatory) and 3 elective subjects

IGCSE O-LEVEL & A-LEVEL

Minimum of 5 credit passes in the IGCSE/O-Levels (Maths and English mandatory) and 3 passes in the A-Levels

ENGLISH IB

Minimum of 5 credit passes in the IGCSE/O-Levels (Maths and English mandatory) and a minimum score of 4 points in 3 Higher Level (HL) subjects

FRENCH IB

Minimum of 50% overall average pass. (Results to be presented to NAB for verification) Maths and English mandatory

AMERICAN HIGH SCHOOL

Minimum GPA of 3.0 (Maths and English mandatory)

COURSE OUTLINE

SEMESTER 1

Course Name

Communication Skills
Data Analysis (using MS Excel)
French Language
Fundamentals of Innovation and Entrepreneurship (FIE) Seminar I
Introduction to Programming with Python
Pre-Calculus (with MATLAB)
Technology and Society

SEMESTER 2

Course Name

Emerging Technologies
Financial Accounting I
Fundamentals of Innovation and Entrepreneurship (FIE) Seminar II
IT Infrastructure Management
Logic and Critical Thinking
Mathematics for Computing (with MATLAB)
Programming in C
Text and Meaning

SEMESTER 3

Course Name

Computer Digital Systems
Data Structures and Algorithms
Database Systems
Fundamentals of Innovation and Entrepreneurship (FIE) I
Global Environment of Business Decisions
Leadership Seminar I
Object Oriented Programming with C++

SEMESTER 4

Course Name

African Studies
Fundamentals of Innovation and Entrepreneurship (FIE) II
Linux Administration
Operating Systems
Probability, Statistics and Reliability (with MATLAB)
Systems Analysis and Design
Programming with Java

SEMESTER 5

Course Name

Computer Networks
Cryptography and Network Security
Introduction to Data Science
Leadership Seminar II
Software Engineering
Storage Management
Web Technologies

SEMESTER 6

Course Name

Cloud Computing
Computer Graphics and Multimedia
Cybersecurity
Industrial Internship
Multimedia Systems
Project Management, Engineering Economics and Risk Analysis
Research Methods in Computing

SEMESTER 7

Course Name

E-commerce

Ethical and Legal Issues in Computing

Human-Computer Interaction

Project Phase I

Technical Elective - I

Technical Elective - II

SEMESTER 8

Course Name

Mobile Computing

Professional Ethics and Values

Project Phase II

Technical Elective - III

Technical Elective - IV

ELECTIVES

Data Science and Cloud Application

Business Intelligence
Cloud Application
Artificial Intelligence
Machine Learning

Cybersecurity and Data Science

Business Intelligence
Security Intelligence
Artificial Intelligence
Machine Learning

Network Security, IoT and Application Development

Mobile Application Development
Security Intelligence
Internet of Things (IoT)
Wireless Networks

Cloud Computing, IoT and Application Development

Cloud Application
Mobile Application Development
Internet of Things (IoT)
Wireless Networks